Lost Wages and House Prices: The Oil Bust Where Norway's Offshore Workers Lived

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Abstract

What does a loss of wages do to a local economy? This paper uses Norway's offshore oil workers and the 2014 oil price fall as a quasi-experiment that separates a wage shock from a shock to local firms. Because these workers spend four weeks at home for every two on the platform, nearly half of them live in municipalities with almost no oil firms. I compare 375 such municipalities in a difference-in-differences design by their share of employed residents who worked offshore in 2003. From 2013 to 2017, income from employment fell by 1.9 percent, employment by 1.4 percent and house prices by 7 percent per percentage point of that share. The housing stock and the population did not move. The oil suppliers in these municipalities lost jobs but do not account for the price fall. House prices absorbed about four times as much of the shock as income did.

JEL: R31, R23, J23, Q33 **Keywords:** house prices, local labour markets, oil price shock, offshore workers, income shocks, Norway

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1 Introduction

The 2014 oil price bust has been studied where the oil industry is located, so a region's workers and its firms are hit together. This paper measures it where the workers lived, in municipalities with almost no oil firms, and asks what lost wages do to a local economy. The bust was large. The oil price had risen from $29 a barrel in 2003 to $109 in 2013, and it fell by more than 70 percent between June 2014 and January 2016. In Norway, investment in oil and gas extraction fell by a quarter in current prices from 2013 to 2016 and exploration investment by more than 40 percent. The number of offshore workers whose workplace was the continental shelf fell by a quarter.

Stavanger, home to the headquarters of the largest operator on the continental shelf, Statoil (Equinor since 2018), and areas close to it diverged from the rest of Norway in income, employment and house prices. Between 2013 and 2016, registered unemployment rose by 2.5 percentage points in the county around Stavanger (Rogaland) and barely changed in the rest of Norway. The average sale price of dwellings rose by about 1 percent in Rogaland and by about 21 percent in the rest of Norway.

Research on the bust follows this contrast. Böjeryd (2025) follows homeowners in the Stavanger region, where the bust lowered earnings and house prices. Aastveit et al. (2025) compare government workers in oil and non-oil regions to measure how spending responded to the fall in house prices. Juelsrud and Wold (2025) compare oil engineers with other skilled workers in the oil region, and Schøne and Strøm (2021) compare couples in labour markets with many and few oil jobs.

Work on resource booms and busts elsewhere also measures them where the industry sits. Oil and gas booms raise local wages and incomes (Allcott and Keniston, 2018; Feyrer et al., 2017) and house prices (Bartik et al., 2019). After a bust, incomes fall below the path they would have followed (Jacobsen and Parker, 2016). Each tradable job carries about 1.6 local service jobs with it (Moretti, 2010), and when household wealth falls the jobs that go are the ones that serve local demand (Mian and Sufi, 2014). Notowidigdo (2020) finds that an adverse local shock pushes fewer people out than a favourable one draws in, and estimates how far house prices and transfers carry the loss for those who stay. In all of these places the workers, the firms and the landowners are hit together. The offshore rotation gives a place where the workers are hit and the oil companies are absent. The fall in prices can then be read as the response to the lost pay once the firm side left in the sample (the equipment suppliers, the yards and the coastal industries) is held fixed.

The places outside the oil regions were not all alike. Few of their local businesses were in the oil industry but some of their residents worked on the continental shelf, in extraction, drilling and the service jobs around them. In 2003, 14,912 Norwegian residents had the continental shelf as their registered workplace. Forty-seven percent of them lived outside Rogaland and Hordaland, the two largest oil counties. They were also spread unevenly.[1] In the 375 municipalities without an onshore oil economy, the share of employed residents working offshore ran from zero in 40 municipalities to more than 3 percent in Tysnes, Fedje, Kvinesdal and Bamble. Over the bust, from 2013 to 2017,

[1] Table 10 lists the ten municipalities with the most offshore work and the ten with the least.

the average sale price of dwellings rose by 38 percent in the half of these municipalities with the fewest offshore workers and by 20 percent in the tenth with the most. The average income from employment rose by 10 percent and 6 percent. Compared with Stavanger all of these places did well but compared with one another, those with more offshore workers did worse.

Offshore workers could live far from the industry because their work leaves them long stretches at home. The Working Environment Act of 1977 was applied to the fixed platforms and limited working hours for round-the-clock work. Since then crews have worked two weeks on the platform followed by at least three weeks ashore (Meland, 2019b). After 1986 the pattern became two weeks on, three off, two on and four off. In 2002 and 2003 it became two weeks on and four off, first by local agreements and then under the offshore collective agreements (Meland, 2019a; Knutsen et al., 2025). The agreements also make the employer pay for the trip between the worker's home and the heliport.[2] A worker who spends most of the year on the mainland does not need to live near the heliport. The worker, in nine cases out of ten a man, and his family can live where they choose and many did so long before the boom. In the census of 1980, 773 (35 percent) of the 2,192 residents at work in the North Sea in the census week lived outside Rogaland and Hordaland. The register series starts in 2000. In every year from then to 2017, between 45 and 50 percent of offshore workers lived outside the two counties (Table 1).

The fall in the oil price that hit Stavanger should therefore have reached these regions too, through the earnings of residents who worked at the shelf. Offshore workers lost their jobs wherever they lived, including in towns with no oil firms at all. This paper asks how a shock that reaches a place only through its residents' wages affects its labour and housing markets. The question is not Norway's alone. Wherever people work far from where they live, a shock to a distant employer reaches their home town through their pay, and rotation work makes that exposure measurable. To answer it, I use public municipal data from Statistics Norway and a difference-in-differences design in which municipalities differ in how much they were exposed. Exposure is the share of a municipality's employed residents who worked on the shelf in 2003, the year before the oil price began to rise, and it is held at that value throughout. By 2003 the rotation had long shaped where offshore workers lived. The sample is the 375 municipalities where at most 0.5 percent of employed residents held onshore oil jobs at the oil companies and their service firms in 2003. I count the jobs in the municipality and the jobs their residents commuted to elsewhere, so the bust could hardly reach them through oil firms, local or nearby. They hold almost half of the offshore workers. The design assumes that, had the oil price not fallen, municipalities with more offshore workers in 2003 would have followed the same path as the others.

From 2013 to 2017, the average income from employment fell by 1.9 percent relative to other municipalities, employment of residents by 1.4 percent and house prices by 7 percent per percentage point of employed residents working offshore in 2003. House prices are measured by the average sale price of dwellings. The dwelling stock and the population did not move during the oil price fall.

[2]The agreement for the mobile rigs has done so since 1974 (Arbeidsretten, 2017), and the operators' agreement of 2024 states it in section 15.1 (Offshore Norge, 2024).

Table 1: Residents whose workplace is the continental shelf, 1980 and 2000–2017

Year	Norway	Rogaland	Hordaland	Elsewhere	Elsewhere (%)
1980 (census week)	2,192	929	490	773	35.3
2000	15,477	4,969	3,605	6,903	44.6
2003	14,912	4,294	3,658	6,960	46.7
2005	16,290	4,842	3,842	7,606	46.7
2010	17,454	5,059	4,154	8,241	47.2
2013	20,603	5,609	4,840	10,154	49.3
2015	17,842	4,767	4,261	8,814	49.4
2017	14,742	4,035	3,624	7,083	48.0

Note: 1980: residents aged 16 and over who were at work in the week of 25–31 October 1980 and whose meeting place was the North Sea, from the full count of the 1980 census (table 9 of Statistics Norway's municipal census booklets). Workers ashore on their rotation that week are not counted, so the 1980 row is comparable with the later rows in its shares, not its levels. A booklet lists the North Sea only when it is among the municipality's main destinations. It is listed by 23 of the 26 municipalities of Rogaland, 19 of the 34 of Hordaland and 69 of the 394 elsewhere, so the municipalities left out lie mostly outside the two counties; the row sums the 110 whose numbers can be read. 2000–2017: employed residents whose workplace is the continental shelf in the fourth quarter, by county of residence on the 2019 map (Statistics Norway table 03321). The register series starts in 2000, and later years are not shown because Statistics Norway stopped recoding offshore establishments to the shelf after 2018. The 1990 census reports the North Sea only from a sample in municipalities with 6,000 or more residents and suppresses its small cells, so it is not shown.

Prices fell about four times as much as income. With the coastal industries and the distance to a heliport held fixed the estimates are about two-fifths smaller, and a straight continuation of the 2009 to 2012 path would then cover zero.[3] The effects sit on the region where offshore workers live rather than on the municipality and the oil firms, the coastal industries and the equipment suppliers left in the sample do not explain them. The loss of income was the men's. Women's income did not fall during the period. For each offshore job lost, the equipment suppliers of the industry lost 1.3 jobs, since the sample rule does not remove them. The rest of the local economy lost about two, although that estimate is imprecise. Whether laid-off offshore workers found other jobs cannot be seen in municipal data, but employment outside oil fell rather than rose, so other industries did not add jobs on net.

Measuring the shock where the workers lived separates a shock to a region's workers from a shock to its firms, and it shows that lost wages move house prices in places with almost no oil firms. The firms that were hit, the equipment suppliers, lost jobs but do not explain the fall in prices.

2 Setting

Much of the oil wealth belongs to the state and goes into a sovereign fund. The fund held about 865 billion dollars at the end of 2014.[4] A fiscal rule lets the government spend about the expected real return of the fund each year, so a fall in the oil price lowers what flows into the fund and not

[3] Table 9.

[4] The fund held 6,431 billion kroner. Dollars are converted at Norges Bank's rate on 31 December 2014 of 7.43 kroner per dollar.

what the budget can spend. The government leaned against the downturn with the budgets of 2015 and 2016, Norges Bank cut the policy rate from 1.5 percent to 0.5 percent between December 2014 and March 2016, and the krone lost about a fifth of its value against the import-weighted index (International Monetary Fund, 2016). These responses reached every municipality, and the comparison across municipalities removes them. The weaker krone helped the coastal exporters, and I hold their industries fixed in the robustness checks.

2.1 Offshore work

Norway's oil and gas lie under the continental shelf, and the onshore oil economy grew up around Stavanger and Bergen (Figure 1). In 2003 the extraction of oil and gas and the services around it employed 28,775 people, 16,593 of them on land.[5] The Stavanger/Sandnes region held 59 percent of the onshore jobs. The other 12,182 jobs were on the shelf itself, which Statistics Norway records as a place of work of its own. With the platform work and catering registered there as well, 14,912 residents had the shelf as their workplace. Crews flew from five heliports, so the workplace fixes the heliport a worker flies from but not where the worker lives. Where they live is not random. Within the sample and weighted by population, the 2003 share correlates -0.53 with the log distance to the nearest heliport and 0.44 with the share working in ship and platform building, and the robustness checks hold these coastal traits fixed.

2.2 Boom and bust

The oil price averaged between 25 and 29 dollars a barrel from 2000 to 2003 and stayed around 110 from 2011 to 2013 (Figure 2). Investment on the shelf rose from 13 billion dollars in 2003 to 37 billion in 2013, at 2013 prices.[6] The number of residents working on the shelf rose from 14,912 in 2003 to 20,603 in 2013. I measure the exposure in 2003, before the boom moved it. The bust began in June 2014 as rising oil production met weaker global demand (Baumeister and Kilian, 2016). The number of residents working on the shelf fell from 20,603 in 2013 to 14,742 in 2017. The cuts fell on exploration, drilling, engineering and supply. These are the activities that employ rotation crews. The loss of offshore pay therefore arrived from 2014 in every municipality that housed offshore workers.

2.3 Layoffs and benefits

A worker who lost an offshore job received the same unemployment insurance as any other worker, 62.4 percent of previous earnings up to a cap of six times the base amount, for up to two years. Employers could also lay workers off temporarily, and the employment register counts a worker on temporary layoff as employed if the layoff is expected to last less than 90 days. The income measure leaves benefits out, so it records the loss of pay before insurance.

[5] Division 11 of the industry classification SN2002.

[6] From 77 to 219 billion kroner at 2013 prices, deflated with the consumer price index. Dollars are converted at Norges Bank's average rate for 2013 of 5.88 kroner per dollar.

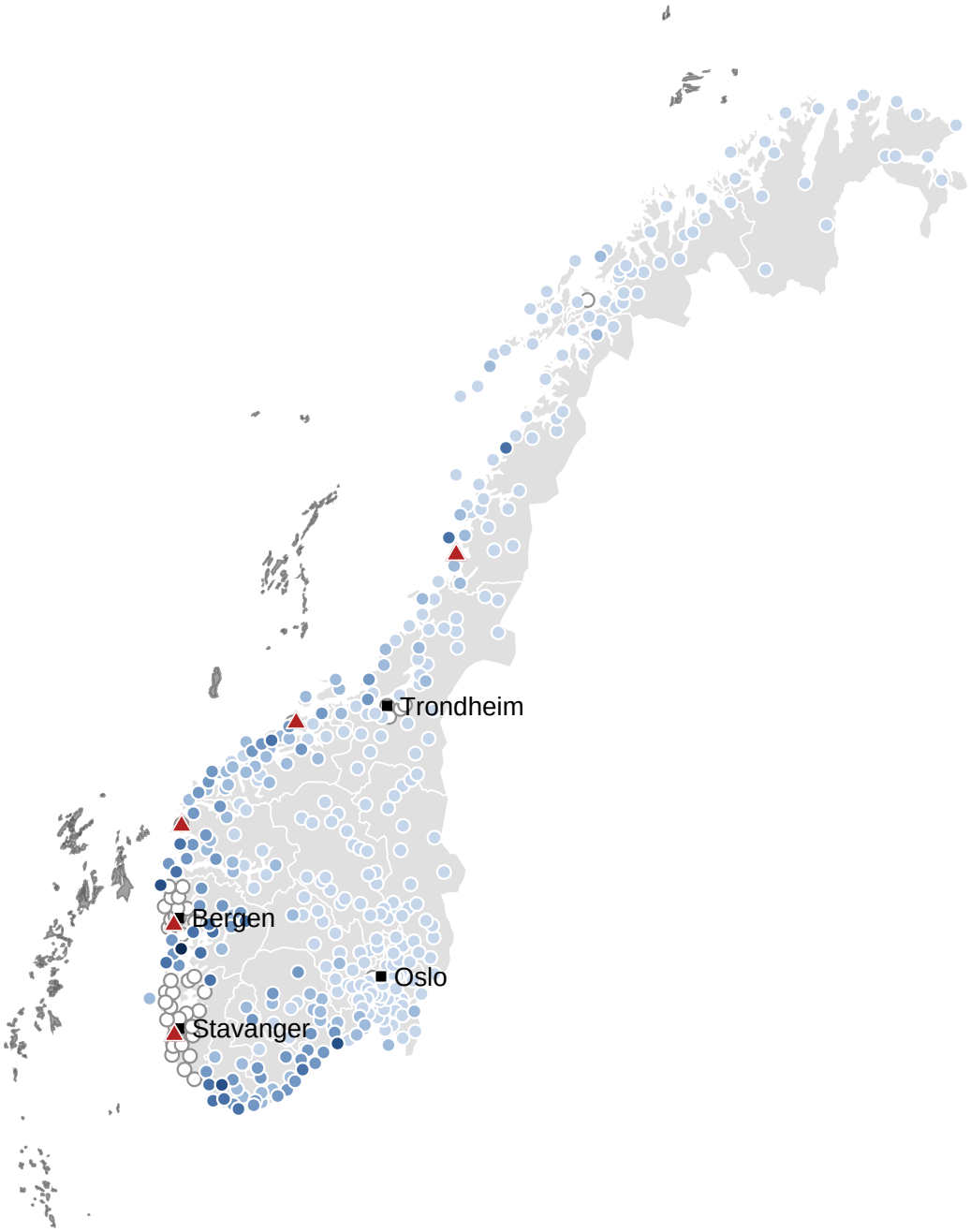


Figure 1: Oil and gas fields, offshore work and the sample

Note: The grey areas at sea are the outlines of the 139 oil and gas fields in the Offshore Directorate's FactMaps in October 2026, which include fields developed after 2003. Triangles mark the five heliports from which crews flew offshore in 2003: Stavanger (Sola), Bergen (Flesland), Florø, Kristiansund and Brønnøysund. Each point on land is a municipality, at its administrative centre. Points in the sample are shaded by the share of employed residents whose workplace is the continental shelf in the fourth quarter of 2003: from lightest to darkest below 0.5 percent, 0.5 to 1, 1 to 2, 2 to 3, 3 to 4, and 4 percent or more. White points are the 46 municipalities outside the sample, where more than 0.5 percent of employed residents held onshore oil jobs, in the municipality or by commuting (Section 3.2). Lines inside Norway are the 18 counties of 2019. Squares mark four towns. Tønsberg and Sandefjord, which are one unit in the data because Stokke was split between them in 2017, are both drawn.



(a) Oil price

(b) Planned and invested petroleum investment

(c) Residents working on the shelf

Figure 2: The boom and the bust, 2000–2019

Note: Panel (a) is the annual mean of the daily Brent spot price in US dollars per barrel (US Energy Information Administration, through FRED). Panel (b) shows accrued investment in oil and gas extraction and pipeline transport (solid line, filled markers; Statistics Norway) and the investment that the oil companies had planned for the same year in the May survey of the year before (dashed line, hollow markers). Both are in billion dollars of 2013, deflated with the consumer price index and converted at Norges Bank's average rate for 2013 of 5.88 kroner per dollar. Panel (c) is the number of residents whose workplace is the continental shelf in the fourth quarter, shown to 2017 (Section 3.1). The dashed lines fall between 2003 and 2004 and between 2013 and 2014.

3 Data

All data used in this paper are retrieved from public sources of Statistics Norway's StatBank, the Labour and Welfare Administration (NAV) and the Norwegian Offshore Directorate.[7] Municipality level data is used in the regressions and each year's data is carried forward to 2019 with Statistics Norway's official list of code changes. If municipalities merged during the sample period, the merged unit carries the whole population throughout the sample. The only split between 2000 and 2019 is that of Stokke in 2017, so I merge its two successors (Tønsberg and Sandefjord) into one unit. This leaves 421 municipalities in 88 economic regions.

3.1 Exposure

Statistics Norway records the continental shelf as a place of work of its own and links each worker to the municipality where they live. For each municipality I divide the number of residents whose workplace is the shelf by the number of employed residents. In 2003, the workplace of 14,912 Norwegian residents was registered as the shelf. Of these, 82 percent worked in the extraction of oil and gas and the services around it[8], 10 percent in platform and yard work and 6 percent in catering. I fix the exposure in 2003, the last year before the oil price began to rise, so exposure is the share of a municipality's employed residents whose workplace was the continental shelf in the fourth quarter of 2003.

The count of offshore workers is a lower bound. Employers register their establishments at fixed addresses, and Statistics Norway moved a job to the shelf only when it knew the establishment

[7] Table 21 in the appendix lists every table.

[8] Division 11 of the industry classification SN2002.

worked offshore. These jobs are mainly in oil services, drilling and catering. Offshore workers of employers registered at an onshore office stay at that office's address, workers on mobile rigs are recorded as seafarers. Foreign residents and commuters from Sweden have no home municipality at all. If every municipality's count misses the same fraction of its offshore workers, they overstate the effect of one true point by that fraction. If the fraction varies across municipalities for reasons unrelated to the outcomes, it weakens them. The share calculated from the 2000 data correlates 0.93 with its 2003 value. The 1980 census counts only the workers who were offshore in the census week and lists the North Sea for a hundred municipalities. With the unlisted municipalities at zero, its share correlates 0.7 with the 2003 share on the sample and 0.8 across all municipalities.[9] Therefore, the shares are assumed to be stable before the boom. The series changes source in 2015 and breaks again in 2018, both well after the year I use to calculate the exposure.

In 2003, the share of employed residents working offshore was zero in 40 municipalities and 0.28 percent at the median (Table 3). It was 1.33 percent at the 90th percentile and above 3 percent in four municipalities (Table 10). Weighted by population, the mean is 0.41 percent and the standard deviation 0.57. The coefficients in this paper are per percentage point of exposure, which is about the difference between the 90th and the 10th percentile. Figure 1 maps the share across Norway.

3.2 Sample

The sample is restricted to municipalities whose residents held almost no onshore oil jobs in 2003 since the design needs places that the bust could reach mainly through their residents. For each municipality I count its employed residents by the municipality where they work and weight each of them by the share of oil extraction and related jobs among all jobs located in that workplace municipality. The sum includes the residents who work in their own municipality, so the measure covers oil firms in the municipality as well as commuting to oil jobs elsewhere. Divided by the municipality's employed residents, it estimates the share of its workers who held onshore oil jobs, on the assumption that a commuter holds an oil job as often as the average worker where they work. Jobs on the continental shelf are left out, since they are the exposure. Across all municipalities the measure allocates the 16,593 onshore oil jobs to the places where their holders lived. The cutoff is set by a gap in the data. In 344 of the 421 municipalities the share was at most 0.2 percent, and in the median one it was 0.07 percent. Above 0.2 percent the values rise without a break to 0.48 percent in Porsgrunn. The next one is Etne with 0.57 percent. I keep the municipalities at or below 0.5 percent. The larger gaps above 1.2 percent would keep the oil-office cities and their commuter suburbs, such as Bærum, Trondheim, Askøy and Karmøy. The rule leaves out the oil centres of Stavanger, Haugesund and Bergen with their commuter belts which corresponds to 46 municipalities. The sample is the remaining 375 municipalities in 86 economic regions. They had 3.8 million inhabitants in 2014, three quarters of the population. They also held 7,134 (48 percent) of the 14,912 offshore workers.

[9] Residents at work in the North Sea in the census week of 1980 per employed resident of 2003, on the 2019 map, weighted by population. The census count is described in the note to Table 1.

A rule based on the oil jobs located in the municipality would not do the same job. It would keep 30 municipalities that this rule leaves out. Most of them are suburbs and islands around Stavanger, Haugesund and Bergen, such as Gjesdal, Rennesøy, Strand, Klepp and Time. They have few oil offices of their own, but their residents commute to the offices next door. In these 30 municipalities, 1.5 percent of employed residents held onshore oil jobs and 2.3 percent worked offshore. In the sample the two shares are 0.1 and 0.41 percent. Including them would put the places most exposed to the onshore oil economy at the top of the offshore distribution.

Division 11 is a narrow definition of the oil jobs. It leaves out the yards, the equipment makers, the supply ships and the supply bases. These were also hit in 2015. I therefore build a second rule for the supply chain, measured in the same year as the exposure. Statistics Norway's list of petroleum-related industries exists only from 2008, five years into the boom. In that year 90 percent of its jobs fell into two divisions of the older classification. These are division 11 and division 35, which covers ships and platforms.[10] Both divisions are published by place of work for 2003 and their share of a region's jobs in 2003 correlates 0.98 with the share on the 2008 list. A second check uses the 2008 list directly. A municipality is a hub if at least 5 percent of the jobs located in it are on the list, or at least 3 percent in a stricter version. Most hubs lie on the coast between Sunnhordland and Trøndelag. Because workers commute within economic regions, the check leaves out every municipality in a region that contains a hub. That leaves 272 of the 375 municipalities at 5 percent and 240 at 3 percent, with 58 and 47 percent of the sample's offshore workers. Table 2 shows this rule and the other samples I use to test the choice of sample, and Section 6 reports the estimates on each.

Table 2: The headline sample and the alternative samples

			Population	Offshore workers		Offshore work (%)	
Sample	Munic.	Regions	(1,000)	Number	% of all	Mean	P90
All municipalities	421	88	5,109	14,912	100	0.67	2.04
Headline sample	375	86	3,849	7,134	48	0.41	1.12
Onshore jobs located in the region only	360	79	3,778	7,358	49	0.44	1.18
Outside Rogaland and Hordaland	362	80	4,144	6,960	47	0.38	0.89
No onshore oil job in the region	259	59	1,721	3,138	21	0.40	0.98
Regional onshore share $\leq 1.2\%$	381	84	4,285	8,211	55	0.43	1.18
Municipal onshore share $\leq 0.5\%$	404	88	4,089	9,513	64	0.53	1.44
Supply chain 2003 $\leq 1\%$	256	56	3,257	3,942	26	0.27	0.77
Supply chain 2003 $\leq 2\%$	296	66	3,506	4,998	34	0.32	0.78

Note: Population on 1 January 2014. Offshore workers are residents whose workplace was the continental shelf in the fourth quarter of 2003, and offshore work is their share of employed residents; the mean and the 90th percentile are weighted by population. The headline rule counts the onshore oil jobs held by a municipality's residents and the regional rule the onshore oil jobs located in its region. The supply-chain rule counts the region's jobs in SN2002 divisions 11 and 35. All three are measured in 2003.

[10] Statistics Norway's correspondence table between SN2007 and SN2002.

3.3 Outcomes

The outcomes follow the shock from income to the labour market and to housing (Table 3). The first stage is the average income from employment per resident aged 17 and over, from the tax records from 2005 onwards. It leaves benefits out, so it measures the loss of pay before insurance and taxes, and because it is measured per resident it moves with both the pay of those in work and the share of adults who work. When municipalities merged, I average their means with population weights. To see whose income fell, I also use it for men and for women separately, since nine in ten of the offshore jobs of 2003 were held by men. As a check I use the median household income after tax, which includes unemployment benefits and so measures the loss after insurance and taxes, averaged with household weights when municipalities merged.

For the labour market I use the number of employed residents aged 20 to 66 in the fourth quarter and the registered unemployment rate in November. The employment register moved to a new source in 2015. Statistics Norway gives no overlap year or back-cast by residence, but it published the 2015 count of employees by workplace municipality twice, on the new source and adjusted to the old one. The gap between the two is the size of the break in each municipality. It does not follow offshore work. I therefore measure employment from 2013 to 2017, like income and prices.[11]

To see which jobs were lost, I also use employed residents by industry of their workplace from 2013 to 2019.

For housing I use the average price per sale of dwellings sold on the free market, the dwelling stock on 1 January and the number of dwellings given a building permit. The average price is total purchase value divided by the number of sales, so I use it only where a municipality had at least 10 sales in every year from 2003 to 2019, which keeps the panel balanced. That leaves 271 of the 375 municipalities, with 96 percent of the sample's population and 95 percent of its offshore workers. The average price is not adjusted for the size or type of the dwellings sold. As a check I also use the price per square metre of detached houses. Statistics Norway publishes it only where there were enough sales, and it is observed in every year in 135 municipalities. Population is the count on 1 January.

The bust windows start in 2013, the last year before the price fell. Income, employment and prices are measured from 2013 to 2017 and registered unemployment from November 2013 to November 2016. The dwelling stock and the population are measured from 1 January 2014 to 1 January 2018. The windows end in 2017 because investment on the shelf and the number of offshore workers reached their lowest level that year and the oil price had begun to recover. The series that counts offshore workers also breaks in 2018, when Statistics Norway stopped recoding offshore establishments to the shelf. The event studies run from 2008 to 2019. For the boom the windows run from 2003 to 2013, or from 2005 for income and from 2006 for the dwelling stock, where those series begin. Every outcome is 100 times the change in logs over its window. Registered unemployment is the change in percentage points.

[11] Table 15 reports the test.

Table 3: Summary statistics, headline sample

			Headline sample				All
Variable	Unit	*N*	Mean	P10	Median	P90	Mean
Exposure and sample rules, 2003							
Offshore work	% employed residents	375	0.41	0.00	0.28	1.33	0.67
Onshore oil economy of the region	% of the region's jobs	375	0.07	0.00	0.00	0.03	0.75
Residents' onshore oil jobs	% employed residents	375	0.10	0.02	0.06	0.19	0.77
Supply chain of the region	% of the region's jobs	375	0.98	0.01	0.35	3.62	2.09
Controls							
Fishing and fish farming	% employed residents	375	0.83	0.00	0.19	8.86	0.72
Ship and platform building	% employed residents	375	1.02	0.05	0.42	3.29	1.37
Water transport	% employed residents	375	0.88	0.08	0.56	3.41	0.97
Distance to the nearest heliport	km	375	261.82	58.29	192.07	445.19	215.76
Population, 1 January 2014	thousands	375	10.26	1.15	4.33	20.60	12.14
Outcomes, bust							
Mean employment income per resident, 2013–17	log points	375	8.76	4.24	9.11	12.89	7.76
Median household income after tax, 2013–17	log points	375	9.71	5.80	9.67	13.43	9.00
Employed residents, 2015–17	log points	375	2.07	−3.24	0.60	3.91	1.80
Registered unemployment, Nov 2013–16	pp	373	−0.11	−1.20	−0.10	1.08	0.26
Average price per sale, 2013–17	log points	271	27.68	8.24	26.27	46.41	24.15
Detached-house price per m^2, 2013–17	log points	225	22.48	5.36	20.45	31.80	18.92
Dwelling stock growth, 2014–18	log points	375	4.66	0.59	3.57	7.38	4.96
Population growth, 2014–18	log points	375	3.42	−3.53	0.74	5.40	3.55

Note: Municipalities in the headline sample (375) and in Norway (421). Means are weighted by population on 1 January 2014, except for population itself; the percentiles are across municipalities. Log points are 100 times the change in logs and pp are percentage points. The average price per sale is shown for the 271 municipalities of the balanced panel, and the detached-house price where Statistics Norway publishes one. The supply chain of the region is jobs located in the region in SN2002 divisions 11 and 35, as a share of all jobs there, in 2003.

3.4 Controls

Every regression holds fixed the size of the municipality and its industry mix before the boom. These are the log population at the start of the window and the 2003 shares of employed residents in industry and mining, in the primary industries and in trade, hotels and restaurants. It also holds fixed two measures of the onshore oil economy. The first is the onshore oil share of the jobs in the municipality's region, since some sample municipalities lie in oil regions. The second is the share of its own residents who held onshore oil jobs, the measure behind the sample rule. Offshore workers live near the heliports and in communities that already worked at sea, so a second set of controls measures those traits in 2003. These are the shares of employed residents in fishing and fish farming, in ship and platform building and in water transport, and the log distance from the municipality's administrative centre to the nearest of the five heliports in use in 2003.

A third set is used in the robustness checks. It holds fixed the distance to the nearest of Oslo, Bergen, Trondheim and Stavanger and the 2008 share of residents working in fabricated metal products and in machinery and equipment, the industries that supplied the oil industry.

4 Empirical strategy

I estimate the effect of the bust as a long difference across municipalities.

$$\Delta y_i = \alpha + \beta\, S_i + W_i'\gamma + \varepsilon_i \tag{1}$$

Δy_i is 100 times the change in the log of the outcome in municipality i over its bust window. The window runs from 2013 to 2017 for income, employment and prices and from 1 January 2014 to 1 January 2018 for the dwelling stock and the population. S_i is offshore work, the share of employed residents whose workplace was the continental shelf in the fourth quarter of 2003, in percentage points. W_i holds six controls measured before the bust. Two describe the onshore oil economy. They are the onshore oil share of the jobs in the municipality's economic region and the share of its employed residents who held onshore oil jobs in 2003. The other four describe the municipality's size and industry mix. They are the log population on 1 January 2014 and the 2003 shares of employed residents in industry and mining outside oil, in the primary industries and in trade, hotels and restaurants. α is a constant and ε_i is the error. The regressions are weighted by population on 1 January 2014, so β is the effect for the average resident per percentage point of offshore work. The robustness checks add the coastal controls and the distance to a large city to W_i.

4.1 Event studies

The event studies make the same comparison year by year.

$$y_{it} = \lambda_t + \sum_{k \neq 2013} \mathbf{1}[t = k]\,(\beta_k\, S_i + W_i'\gamma_k) + \varepsilon_{it}, \qquad y_{i,2013} = 0 \tag{2}$$

The outcome y_{it} is the change in municipality i from 2013 to year t, so it is zero in 2013. For income, prices and the dwelling stock it is 100 times the change in logs, and for employment it is the change in the employment rate in percentage points. λ_t is a year effect, and $\mathbf{1}[t = k]$ equals one in year k. The coefficient β_k on offshore work in year k is therefore the cumulative effect of the bust up to year k after 2013. Before 2013 it tests whether exposed municipalities were already diverging from the others. W_i holds the controls of Equation (1) with the log population of 1 January 2004 in place of 2014, and each control has its own coefficient γ_k in every year. ε_{it} is the error. The regressions are weighted by population on 1 January 2004. The panel covers 2003 to 2019 for prices, 2005 to 2019 for income, 2005 to 2018 for the dwelling stock and 2008 to 2019 for the employment rate.

4.2 Region and municipality

To see whether the effect sits on the municipality or on its region, I split offshore work into two parts.

$$\Delta y_i = \alpha + \beta_R\, \bar{S}_r + \beta_W\, (S_i - \bar{S}_r) + W_i'\gamma + \varepsilon_i \tag{3}$$

$\bar{S}_r$ is the mean of S over the sample municipalities of economic region r, the region of municipality i, weighted by population on 1 January 2004. $S_i - \bar{S}_r$ is the municipality's deviation from that mean. β_R is the effect of living in a region with more offshore workers and β_W the extra effect of having more of them than the neighbouring municipalities. Δy_i, W_i, the weights and the error are those of Equation (1). With a fixed effect for each economic region in place of $\bar{S}_r$ and of the regional onshore share, only the variation within regions is left, and the coefficient on S_i estimates β_W.

4.3 Inference

Standard errors are clustered by the 86 economic regions of the sample. Every p-value and star comes from a wild cluster bootstrap-t over regions with 9,999 draws (Cameron et al., 2008; Roodman et al., 2019). The bootstrap re-estimates the model under the null and perturbs the residuals of each region by a random weight, which gives tests of the right size when clusters are few or unequal in size, and the six-point weights of Webb (2023) are the ones designed for that case. The main table also reports Romano–Wolf p-values for the six bust outcomes tested together (Romano and Wolf, 2005, 2016; Clarke et al., 2020). They control the chance that any one of the six rejections is false, by resampling the whole family in one step-down procedure.

4.4 Identification

The identifying assumption is that, given the controls, municipalities with more offshore workers in 2003 would have followed the same path after 2013 as the others had the oil price not fallen. With one national shock, identification comes from the shares (Goldsmith-Pinkham et al., 2020).

Four threats matter. The first is the oil industry left in the sample. The sample rule caps the onshore oil jobs held by residents at 0.5 percent, so little of that industry remains. The two onshore measures hold what remains fixed. The equipment suppliers and yards that the rule does not remove are held fixed in turn.[12] The second is coastal geography and the krone, which lost a fifth of its value in the bust and helped the coastal exporters. The coastal controls hold the fishing, shipbuilding and shipping shares and the heliport distance fixed, and so address both. The third is that peripheral places follow the national housing cycle less closely, and the fourth is that the only years before the bust are boom years, so a trend before 2013 would be hard to tell from the bust. The event studies show no such trend, and the 2003 share predicts none of the outcomes before the boom.[13]

5 Results

Table 4 reports the bust under six specifications, from no controls to economic-region fixed effects. In the baseline of column (3), from 2013 to 2017 the average income from employment fell by 1.9 log

[12] Table 6.

[13] Table 16 regresses population growth in the 1980s and 1990s and the change in employment and registered unemployment from 2000 to 2003 on the 2003 share, on the headline sample and on the samples of the robustness checks. One of the 24 coefficients has a p-value of 0.05.

points per percentage point of offshore work, employment of residents by 1.4 and the average sale price of dwellings by 7.1. The three effects survive the Romano–Wolf correction for six outcomes. The dwelling stock did not move in any column. Population growth did not move in the baseline. With the coastal controls it is lower, but it was already falling at least as fast from 2009 to 2013, before the bust. A municipality at the 90th percentile of exposure has 1.3 points more offshore workers than one at the 10th, so the baseline implies an income about 2.6 percent lower and prices about 9.5 log points lower in the more exposed of the two. Without controls the effects are larger because the places with offshore workers are small and rural. They are 2.3 for income, 3.0 for employment and 7.7 for the price. The log population and the 2003 industry shares absorb that, and the two onshore measures change little. The coastal controls take about two-fifths from the income effect and a quarter from the price effect and nothing from employment. Adding the distance to a large city takes a further seventh from the price and a quarter from employment. Region fixed effects remove the price effect and most of the rest, which Section 5.1 takes up. Column (7) keeps the 46 municipalities with an onshore oil economy and relies on the onshore controls instead of the sample rule. The income effect is about the same at 1.7 per point, while employment falls by 0.8 and prices by 5.0. Within the oil regions the regional onshore share absorbs the regional fall in prices. What is left there is the variation in offshore work between neighbouring municipalities, and that variation does not move prices.[14] Offshore work in 2013 was 1.13 times its 2003 level and correlates 0.93 with it, so per point of exposure at the time of the bust the effects are about a tenth smaller.[15]

Figure 3 shows the timing. Income, employment and prices are flat relative to 2013 from 2009 to 2012. Prices start to fall in 2014 and keep falling through 2019, employment falls in 2015 and 2016, and income follows from 2015. All three are still down in 2019.

5.1 Regional effect

Four-fifths of the variation in exposure lies between economic regions, and so do the effects (Table 5). The price effect sits on the region's mean exposure, at -9.9 log points per point, and a municipality with more offshore workers than its neighbours shows no larger fall. Column (6) of Table 4 replaces the region mean by region fixed effects and leaves only the within-region part. That part is -0.7 for income, -0.7 for employment and 1.6 for the price. None of the three can be told from zero at 5 percent. The income effect is mostly regional too, with a within-region part about a third as large.

5.2 Supply chain and coast

The oil industry left in the sample does not explain the regional effect (Table 6). The price effect stays between -6.7 and -10.6 log points when the coastal industries, the heliport distance, the supply-chain jobs, the equipment suppliers and the distance to Stavanger and Bergen are held fixed. It stays in that range when the regions with a supply-chain hub are left out. The income effect loses

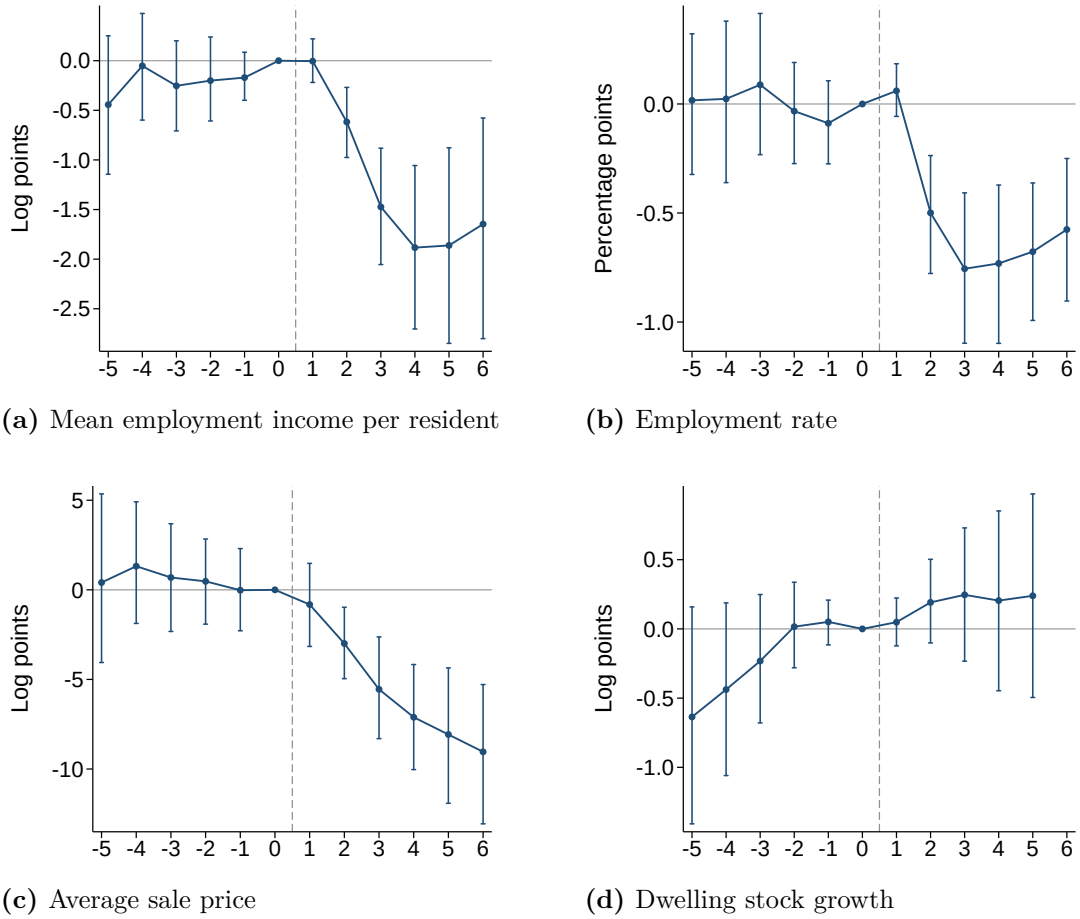
[14] On all 421 municipalities with offshore work interacted with a dummy for the 46, the income effect is -1.8 in the 375 and -1.6 in the 46. The price effect is -7.6 and -1.5, and the difference has a p-value of 0.009.

[15] Table 19.

Table 4: The bust where offshore workers live, 2013–2017

	No controls (1)	Onshore oil (2)	Baseline (3)	Coastal (4)	Distance (5)	Region FE (6)	All 421 (7)
Panel A. Mean employment income							
Offshore work, 2003	-2.293^{***}	-2.340^{***}	-1.932^{***}	-1.200^{***}	-1.139^{***}	-0.717^{*}	-1.708^{***}
	(0.370)	(0.385)	(0.385)	(0.372)	(0.346)	(0.410)	(0.259)
Adjusted R^2	0.25	0.25	0.35	0.45	0.45	0.63	0.67
Observations	375	375	375	375	375	375	421
Panel B. Employment							
Offshore work, 2003	-2.998^{***}	-3.538^{***}	-1.378^{***}	-1.355^{***}	-1.056^{***}	-0.705	-0.823^{***}
	(0.705)	(0.653)	(0.386)	(0.399)	(0.359)	(0.523)	(0.297)
Adjusted R^2	0.19	0.27	0.55	0.56	0.57	0.68	0.54
Observations	375	375	375	375	375	375	421
Panel C. Average sale price							
Offshore work, 2003	-7.718^{***}	-7.725^{***}	-7.142^{***}	-5.158^{**}	-4.127^{**}	1.643	-5.040^{***}
	(1.117)	(1.295)	(1.444)	(1.726)	(1.699)	(2.999)	(1.116)
Adjusted R^2	0.12	0.12	0.13	0.16	0.18	0.31	0.34
Observations	271	271	271	271	271	271	312
Panel D. Dwelling stock growth							
Offshore work, 2003	-0.341	-0.245	0.226	-0.287	-0.310	0.035	0.453
	(0.375)	(0.399)	(0.302)	(0.297)	(0.317)	(0.501)	(0.315)
Adjusted R^2	0.00	0.01	0.12	0.15	0.15	0.37	0.12
Observations	375	375	375	375	375	375	421
Panel E. Population growth							
Offshore work, 2003	-1.505^{**}	-1.893^{***}	-0.284	-0.734^{**}	-0.515^{*}	-0.389	0.016
	(0.491)	(0.438)	(0.272)	(0.296)	(0.279)	(0.327)	(0.264)
Adjusted R^2	0.08	0.16	0.47	0.49	0.50	0.65	0.40
Observations	375	375	375	375	375	375	421
Onshore oil controls		✓	✓	✓	✓	✓	✓
Population and industry controls			✓	✓	✓	✓	✓
Coastal controls				✓	✓		
Distance to a large city					✓		
Economic-region fixed effects						✓	
Onshore oil municipalities kept							✓
Population weights	✓	✓	✓	✓	✓	✓	✓

Note: Coefficients per percentage point of offshore work in 2003 in the 375 municipalities of the headline sample (Section 3.2); the price is on the 271 municipalities of the balanced panel. The outcomes are the bust long differences of Section 3.3, and employment runs across the 2015 change of register source (Section 6.3). Column (2) adds the two onshore oil measures of Section 3.4 and column (3) the log population and the 2003 industry shares. Column (4) adds the coastal controls and column (5) the distance to the nearest large city. Column (6) keeps the controls of column (3) but replaces the regional onshore measure by economic-region fixed effects. Standard errors clustered by economic region are in parentheses. Stars come from the region wild bootstrap with * $p < 0.10$, ** $p < 0.05$ and *** $p < 0.01$. In column (3) the Romano–Wolf p-values are 0.002, 0.017, 0.002, 0.553 and 0.553. They correct for six outcomes, these five and the change in registered unemployment. Column (7) keeps the 46 municipalities that the sample rule leaves out, with the controls of column (3); its price is on the 312 municipalities of the balanced panel. Each municipality is one observation.



(a) Mean employment income per resident

(b) Employment rate

(c) Average sale price

(d) Dwelling stock growth

Figure 3: Offshore work, income, employment and housing, 2008 to 2019

Note: Coefficients β_k of Equation (2) per percentage point of offshore work in 2003, with 95 percent region wild bootstrap intervals, in the 375 municipalities of Table 4. The horizontal axis counts years from 2013, so −5 is 2008 and 6 is 2019, and the dashed line falls between 2013 and 2014. Income is 100 times the log difference from 2013 of the average income from employment per resident. The employment rate is the change from 2013 in percentage points and changes source in 2015. The average sale price is on the 271 municipalities of the balanced panel, and dwelling stock growth is 100 times the log difference of the end-of-year stock from 2013, which ends with the stock of 1 January 2019.

about a third with the coastal controls and holds in the other columns except with the distance quintiles. Leaving out the regions with a 3 percent hub drops 135 municipalities. The price and income coefficients hardly move but their standard errors grow by a quarter to a half, so what is lost is precision. The employment coefficient halves, and that effect does not survive the stricter exclusion.

5.3 Who lost

The income loss fell on men, who held nine in ten of the offshore jobs. Income per adult man fell by 3.1 log points per point of exposure and per adult woman by 0.0, which cannot be told from zero. For men the loss splits between fewer men in work and lower pay per man in work. By 2017 the first had fallen by 1.2 log points and the second by 1.8. The offshore jobs actually lost were 0.35

Table 5: Region mean and own exposure, bust 2013–2017

Dependent variable	Income (1)	Employment (2)	Price (3)
Offshore work, region mean	−2.342***	−1.633***	−9.870***
	(0.443)	(0.420)	(1.622)
Offshore work, own deviation	−0.625*	−0.565	2.034
	(0.324)	(0.468)	(2.339)
Adjusted R^2	0.37	0.55	0.17
Observations	375	375	271
Onshore oil controls	✓	✓	✓
Population and industry controls	✓	✓	✓
Population weights	✓	✓	✓

Note: The table splits offshore work into the region mean, the population-weighted mean over the region's sample municipalities (weights from 1 January 2004), and the municipality's deviation from it (Equation (3)). Column (6) of Table 4 replaces the region mean by economic-region fixed effects. Outcomes, controls, weights, inference and stars as in column (3) of Table 4.

per 100 employed residents per point, while total employment fell by 1.4. No industry outside oil gained except agriculture, forestry and fishing. Not all of the rest is a multiplier. The equipment suppliers of the industry are machinery and metal fabrication and the support codes on Statistics Norway's petroleum list. They lost 0.45 jobs per 100, or 1.3 for each offshore job. That is a direct hit on local firms, because the sample rule removes the oil companies and their service firms but not their suppliers. The rest of the local economy lost 0.64 jobs per 100 ($p = 0.10$), mainly in hotels, restaurants and transport. That is 1.9 for each offshore job, with a 95 percent interval from −0.1 to 3.8. Removing all of manufacturing instead leaves 1.5. The multiplier on offshore pay is therefore about two, not the three to four that the raw count outside oil suggests, and it is not precisely estimated. Dropping the regions with yards and other supply-chain jobs does not remove the suppliers' loss. On the two supply-chain samples of Table 2 it is larger than on the headline sample, 0.50 and 0.64 jobs per 100, because the firms that lost the jobs make machinery and metal parts and sit outside the yard regions. In the rest of the local economy the loss is 0.9 and 0.5 jobs per 100 with intervals that include zero. That is three and one and a half times the offshore jobs.[16] Valued at the pay of the oil industry, the offshore jobs lost explain about a quarter of the fall in employment income.[17]

Households were compensated for part of the loss. Per point of exposure, the average income from employment per adult fell by 1.9 log points. Gross income per adult fell by 1.4. It adds

[16] Tables 11 and 12 report the change in employed residents by industry of workplace, by industry group and for the petroleum-related industries and the manufacturing divisions. Table 13 reports the decomposition and the ratios, and Table 14 repeats them on the supply-chain samples.

[17] The offshore jobs lost are 0.35 per 100 employed residents per point of exposure. Earnings per job in extraction and its services were 1.57 times the all-industry average in the fourth quarter of 2016 (Statistics Norway). The direct loss is therefore $0.35 \times 1.57 = 0.55$ percent of aggregate employment earnings, a share of 0.28 of the fall of 1.9 percent in employment income per adult. It is a lower bound, because it counts every lost job as a full year of pay lost and ignores pay cuts for those who kept their jobs.

Table 6: The regional effect against oil-industry and coastal controls, bust 2013–2017

	Baseline (1)	Coastal (2)	Horse race (3)	Hubs 5% (4)	Hubs 3% (5)	Distance (6)	Suppliers (7)
Panel A. Mean employment income							
Offshore work, region mean	−2.342***	−1.482***	−1.949***	−1.997***	−1.660**	−1.260	−1.538***
	(0.443)	(0.443)	(0.559)	(0.548)	(0.534)	(0.600)	(0.406)
Adjusted R^2	0.37	0.46	0.46	0.41	0.38	0.47	0.53
Observations	375	375	375	272	240	375	375
Panel B. Employment, 2013–17							
Offshore work, region mean	−1.633***	−1.612***	−1.762***	−1.388*	−0.704	−1.317**	−1.573***
	(0.420)	(0.455)	(0.554)	(0.755)	(0.691)	(0.423)	(0.453)
Adjusted R^2	0.55	0.56	0.56	0.53	0.54	0.58	0.56
Observations	375	375	375	272	240	375	375
Panel C. Price							
Offshore work, region mean	−9.870***	−7.882***	−10.552***	−8.952**	−7.997*	−6.704**	−7.890***
	(1.622)	(1.935)	(2.043)	(2.531)	(2.958)	(2.378)	(1.922)
Adjusted R^2	0.17	0.18	0.19	0.19	0.19	0.18	0.21
Observations	271	271	271	197	176	271	271
Coastal controls		✓	✓	✓	✓	✓	✓
Other oil measures			✓				
Hub regions left out				✓	✓		
Distance quintiles						✓	
Supplier exposure							✓

Note: Coefficients on the region mean of offshore work in 2003, in percentage points of employed residents. Every regression includes the municipality's deviation from the region mean and the controls, weights and inference of column (3) of Table 4. Column (1) is the first row of Table 5. Column (2) adds the coastal controls of Section 3.4 and the region's share of petroleum-related jobs in 2008. Column (3) also adds the share of petroleum-related jobs located in the municipality in 2008 and the share of onshore oil workers among its residents in 2003, each as the municipal value and the region mean. Columns (4) and (5) use the controls of column (2) and leave out the regions that contain a hub, a municipality with at least 5 or 3 percent of its jobs on Statistics Norway's petroleum-related list. Column (6) adds four dummies for quintiles of the distance to the nearer of Stavanger and Bergen. Column (7) adds the 2008 supplier share of Section 3.4, as the municipal value and the region mean. Stars as in Table 4.

benefits, pensions and capital income for the same people. Transfers and other income therefore replaced about a quarter of the lost pay. The median household income after tax fell by 1.1. An offshore worker who lost the job received 62.4 percent of previous earnings for up to two years (Section 2.3), and the benefit records show it. Per point of exposure, the number of residents who received unemployment benefits during the year was 0.5 per 100 residents higher in 2016 and 2017 than in other municipalities and 0.2 higher in 2018. That is about two recipients for every three jobs lost. The benefits paid were about 43 dollars per resident higher in 2016, less than a tenth of the lost pay.[18] The oil jobs of residents also partly came back in 2018 and 2019. Waiting at home was therefore a reasonable choice, which fits the unchanged population and the small net movement into other jobs. Prices fell all the same, so the housing market priced the loss of pay and not the insured income.

[18] 360 kroner, converted at Norges Bank's average rate for 2016 of 8.40 kroner per dollar.

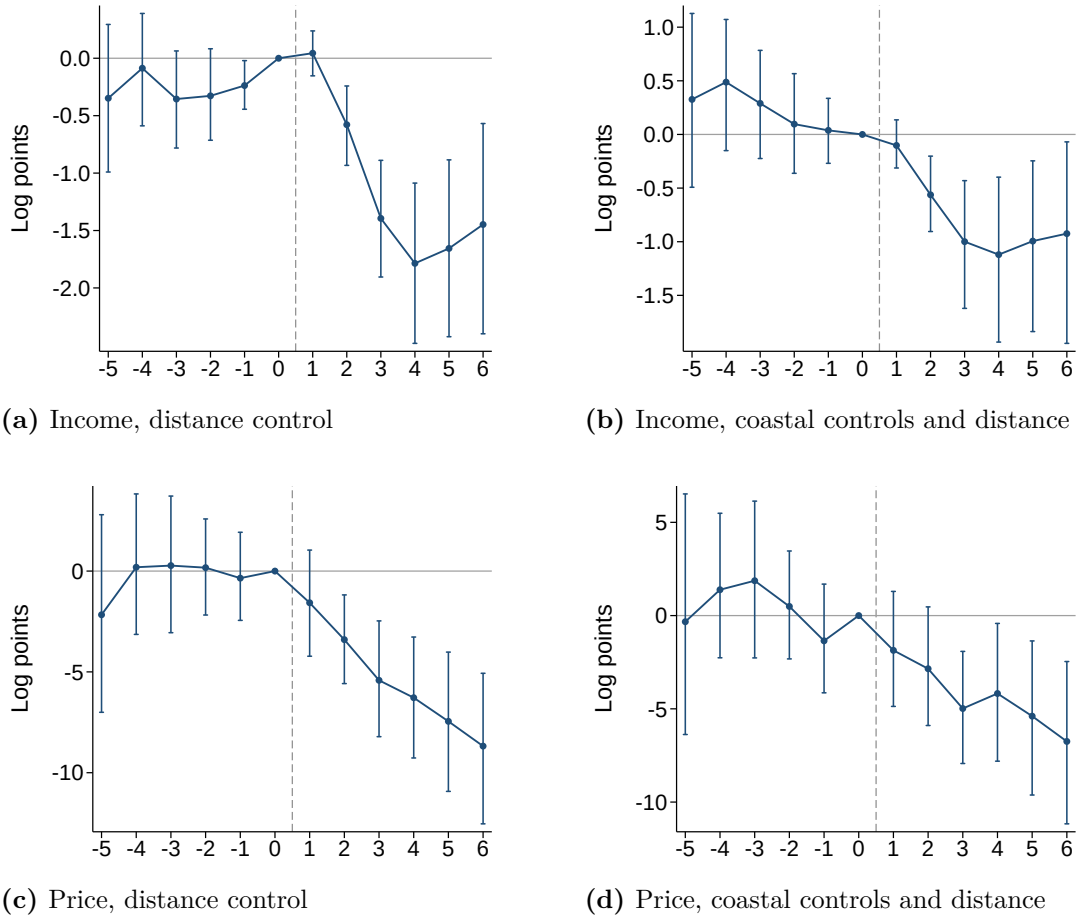
6 Robustness

Table 7 adds the distance to the nearest large city, which costs about an eighth of the price effect and a twentieth of the income effect. Adding the coastal controls as well leaves a price effect of −4.1 and an income effect of −1.1. The median household income after tax includes benefits and gives −1.1 against −1.9 for the mean. The price per square metre of detached houses gives larger effects on its smaller balanced panel. One observation per region gives the same effects as the region mean in Table 5. Figure 4 shows that the paths before 2013 stay flat with these controls.

Table 7: The bust with rural controls and other measures of income and price, 2013–2017

Dependent variable	Income (1)	Employment (2)	Price (3)
Panel A. Added controls			
Baseline	−1.932***	−1.378***	−7.142***
	(0.385)	(0.386)	(1.444)
Distance to the nearest large city	−1.824***	−1.106***	−6.219***
	(0.328)	(0.345)	(1.450)
Coastal controls and distance	−1.139***	−1.056***	−4.127**
	(0.346)	(0.359)	(1.699)
Panel B. Other measures of income and price			
Median household income after tax	−1.088***	–	–
	(0.218)		
Observations	375		
Detached price per m^2, balanced 2003–19	–	–	−10.285***
			(1.921)
Observations			135
Detached price per m^2, every municipality with a price	–	–	−8.302***
			(1.438)
Observations			225
Average price per sale, every municipality with 10 sales	–	–	−6.852***
			(1.428)
Observations			313
Panel C. One observation per economic region			
Offshore work, region mean	−2.310***	−1.353**	−10.241***
	(0.513)	(0.468)	(1.944)
Regions	86	86	86

Note: Coefficients per percentage point of offshore work in 2003 with the specification, weights and inference of column (3) of Table 4. Panel A adds to the baseline the distance to the nearest large city and then the coastal controls of Section 3.4. Panel B uses the other income and price measures of Section 3.3: the median household income after tax, the detached-house price per square metre on the municipalities with a price in every year from 2003 to 2019 and on every municipality with a price in 2013 and 2017, and the average price per sale on every municipality with at least 10 sales in 2013 and 2017. In panel C each economic region is one observation. Its outcome is the population-weighted mean over its sample municipalities. It is regressed on the region mean of offshore work, the region's onshore oil job share, log population and the region means of the 2003 industry shares. The regressions are weighted by region population, with heteroskedasticity-robust standard errors and the wild bootstrap.



(a) Income, distance control

(b) Income, coastal controls and distance

(c) Price, distance control

(d) Price, coastal controls and distance

Figure 4: Income and price with the distance and coastal controls, 2008 to 2019

Note: The event studies of Figure 3 for the mean employment income per resident and the average sale price, with the controls of the second and third rows of panel A of Table 7 added, each interacted with year. The horizontal axis counts years from 2013. Points, intervals, weights and samples as in Figure 3.

6.1 The boom and the national cycle

Two of the threats of Section 4.4 concern the years before the bust, and Table 8 answers both. If the bust were the mirror image of the boom, one coefficient on national offshore employment would describe each outcome over the whole cycle. For income it does not. Income rose with offshore employment in the early boom and fell with it in the bust, but it did not move from 2009 to 2012 while offshore employment kept growing, and the proportional path is rejected with a p-value of 0.002. Prices did not follow offshore employment in the boom at all, so the fall after 2014 is not the late boom unwinding. The second threat is that exposed places follow the national house-price cycle less closely. They do, with a coefficient of -0.03 on the national index. Holding the national cycle fixed makes the bust price effect larger, -8.4 instead of -6.6. The national cycle therefore does not produce the fall.

Table 8: One coefficient on national offshore employment, 2003–2017

	Mean employment income		Average sale price	
	(1)	(2)	(3)	(4)
Whole cycle	-1.45^{***}	-1.43^{***}	-3.37^{**}	-4.37^{***}
2003–2008	-1.88^{***}	-0.40	-3.04	2.75
2009–2012	-0.30	0.53	1.28	4.11^{**}
2014 onward	-1.70^{***}	-2.22^{***}	-6.57^{***}	-8.36^{***}
Offshore work × national house prices		0.001		-0.032^{*}
Proportional path, p-value	0.002	0.002	0.059	0.056
National house-price term		✓		✓

Note: Coefficients β of Equation (4), per percentage point of offshore work in 2003 and per unit of the 2013–2017 fall in national offshore employment, from the event-study sample of Figure 3 for 2003 (income 2005) to 2017. The rows for periods estimate β separately for 2003–2008, 2009–2012 and 2014–2017 in one regression. Columns (2) and (4) add offshore work times the national house-price index (κ). The last row is the p-value of the joint test that the event-study coefficients equal βZ_t. Controls, weights and inference as in Figure 3.

$$y_{it} = \lambda_t + \beta\, S_i Z_t + \kappa\, S_i h_t + \sum_{k \neq 2013} \mathbf{1}[t = k]\, W_i' \gamma_k + \varepsilon_{it},$$
$$Z_t = \frac{N_{2013} - N_t}{N_{2013} - N_{2017}} \tag{4}$$

N_t is the number of residents working offshore in year t, so Z_t is the national shortfall of offshore work below its 2013 level, in units of its fall from 2013 to 2017. It is zero in 2013, one in 2017 and positive in the boom years, when fewer residents worked offshore than in 2013. h_t is 100 times the log of the national price index for existing dwellings relative to 2013, and κ measures how much less closely exposed places follow it. I also estimate β separately for 2003 to 2008, 2009 to 2012 and the bust, and test whether the year-by-year coefficients of Equation (2) equal βZ_t. β is therefore the effect per point of offshore work of a fall in national offshore employment as large as that from 2013 to 2017. λ_t, the controls W_i with their yearly coefficients γ_k and the weights are those of Equation (2), the panel runs to 2017, and ε_{it} is the error.

6.2 Pre-trends

Table 9 asks how far the path of 2009 to 2012 could continue after 2013 before the 2017 effect loses significance (Rambachan and Roth, 2023). With the baseline controls both the price and the income effect survive a straight continuation of the trend. With the coastal controls they do not.

6.3 Further checks

Six further checks are in the appendix.[19] Without population weights the main effects are -1.3 for income, -0.7 for employment and -5.6 for the price, each with a bootstrap p-value below 0.02. The dwelling stock and the population stay at zero. In the 256 municipalities whose region had no

[19] Tables 17 to 20 and Figures 5 and 6.

Table 9: Sensitivity of the 2017 effect to the trend before 2013

	Average sale price		Mean employment income	
	Baseline (1)	Coastal (2)	Baseline (3)	Coastal (4)
Coefficient, 2017	−7.11	−5.16	−1.88	−1.18
95% interval	[−9.9, −4.3]	[−8.5, −1.8]	[−2.6, −1.2]	[−1.9, −0.5]
95% interval, 2009–13 trend continues ($M = 0$)	[−9.6, −1.8]	[−7.1, 3.5]	[−2.7, −1.1]	[−1.5, 0.3]
Breakdown M, smoothness	0.25	0.00	0.25	0.00
Breakdown $\bar{M}$, relative magnitudes	0.75	0.25	1.25	0.75

Note: Bounds of Rambachan and Roth (2023) for the 2017 coefficient of Equation (2), using the 2009–2012 coefficients as the pre-period. The smoothness restriction lets the 2009–2013 trend continue after 2013, with its slope changing by at most M a year; $M = 0$ is a straight continuation. The relative-magnitudes restriction bounds each yearly violation after 2013 by $\bar{M}$ times the largest yearly movement in 2009–2013. The breakdown value is the smallest M or $\bar{M}$ on the grid at which the 95 percent interval contains zero. The covariance is the cluster-robust one by economic region. Coastal adds the controls of column (2) of Table 6.

onshore oil job in 2003, the effects are about half as large, −1.2 for income and −4.6 for the price. They do not differ significantly from those in the regions with some onshore oil. Those regions also hold most of the high offshore shares, so the data cannot separate a residual onshore channel from a dose that bites harder at high levels.[20] The average sale price shows no trend from 1992 to 2002 with the baseline controls and a borderline one with the coastal controls, with joint p-values of 0.30 and 0.05. The rise of prices in exposed places in 2008 and 2009 goes with a crisis cushion in men's earnings, which held up in those two years relative to the surrounding ones, and it shows in the detached-house index rather than in the average sale price. The 2013 share by least squares gives smaller effects than the instrumented one, 1.4 against 1.7 log points for income, 0.8 against 1.2 for employment and 5.2 against 6.4 for the price. Registered unemployment rises in 2015 and 2016 but does not survive the coastal controls, and the centrality classifications of Statistics Norway give the same results as the distance to the nearest large city. The 2015 change of source in the employment register is a level shift of about 3 percent that does not follow offshore work. Statistics Norway published the 2015 count of employees by workplace municipality both on the new source and adjusted to the old one. The gap between the two is unrelated to exposure. This holds in total, outside mining and manufacturing, and in public administration, education and health. On the adjusted series, jobs located in exposed municipalities fell by 0.8 per point of exposure from 2014 to 2015 and by 1.0 on the unadjusted series. The fall in the year of the break is therefore the shock and not the break.[21] The fall in prices did not come with more sales. Per point of offshore work, the number of free-market sales rose by 3 log points from 2013 to 2014 and by 2 from 2013 to 2017.

[20] On the headline sample with offshore work interacted with a dummy for regions with some onshore oil job, the income effect is −1.17 without and −2.11 with ($p = 0.21$ for the difference), employment −0.52 and −1.75 ($p = 0.12$) and the price −4.57 and −8.16 ($p = 0.26$). The 90th percentile of offshore work is 1.04 percent in the regions without onshore oil and 1.80 percent in the others.

[21] Table 15. The adjusted count carries 2014 forward with the change from the first quarter of 2015 to the first quarter of 2016 by industry group and workplace. It is therefore a bridge and not an observation. It counts employees only.

Neither change can be told from zero. Sales per 1,000 dwellings moved by less than 1.5 in any year. A wave of forced sales would raise the number of transactions, and it does not appear, so prices fell on an unchanged volume.[22]

7 Conclusion

The 2014 oil price bust is usually read off the oil regions, where the firms, the workers and the landowners were hit together. Norway's offshore rotation separates them. Offshore workers lived all over the country, and the municipalities with almost no oil firms received the bust as lost pay. Across 375 such municipalities, each percentage point of employed residents working offshore in 2003 cost about 2 percent of employment income and 7 percent of house prices between 2013 and 2017, while the dwelling stock and the population did not move. The loss fell on men. The oil suppliers in these places lost about one job for each offshore job and the rest of the local economy about two, and neither the suppliers nor the rest of the oil industry left in the sample explains the fall in prices.

Two things follow. A shock that arrives through wages is enough to move house prices, and it moves them far more than it moves income. Prices fell about four times as much as income per point of exposure. Why buyers discounted the loss so heavily, when insurance replaced part of it and the oil jobs of residents partly came back by 2019, is the question the paper leaves open. The local multiplier on offshore pay is about two once the suppliers that were hit directly are set aside, close to the multipliers found for tradable jobs elsewhere, although it is not precisely estimated.

The estimates have limits that the paper states. The effects sit on the economic region rather than the municipality, and the coastal traits of the places where offshore workers live account for about two-fifths of them, so the reading rests on the controls that hold those traits fixed. Public municipal data cannot follow a laid-off worker to a new job or to the sale of a house. Register data on individuals would show whether the workers who stayed took lower-paid jobs or waited on benefits, and whether the dwellings that were sold were theirs. Until then the municipal aggregates give the size of the response. Where the industry is absent, its wages still set the price of houses.

References

Knut Are Aastveit, Jesper Böjeryd, Magnus A. H. Gulbrandsen, Ragnar Enger Juelsrud, and Kasper Roszbach. What do 12 billion card transactions say about house prices and consumption? Working Paper 15/2025, Norges Bank, 2025.

Hunt Allcott and Daniel Keniston. Dutch disease or agglomeration? The local economic effects of natural resource booms in modern America. *The Review of Economic Studies*, 85(2):695–731, April 2018. doi: 10.1093/restud/rdx042. URL `https://doi.org/10.1093/restud/rdx042`.

Arbeidsretten. Judgment AR-2017-24, case 10/2017, on the term "fremmøtested" in the collective agreement for mobile offshore installations, section 3.3. Labour Court of Norway, 2 October 2017, 2017. URL `https://www.arbeidsretten.no/assets/Arkiv/ar-2017-24-sak-10-2017.pdf`.

[22] Figure 7. The sample is the 269 municipalities with at least 20 free-market sales in 2013, and the controls, weights and inference are those of column (3) of Table 4. The count mixes forced and voluntary sales, and listings are not published by municipality.

Alexander W. Bartik, Janet Currie, Michael Greenstone, and Christopher R. Knittel. The local economic and welfare consequences of hydraulic fracturing. *American Economic Journal: Applied Economics*, 11(4): 105–155, 2019.

Christiane Baumeister and Lutz Kilian. Understanding the decline in the price of oil since June 2014. *Journal of the Association of Environmental and Resource Economists*, 3(1):131–158, 2016. doi: 10.1086/684160. URL `https://doi.org/10.1086/684160`.

Jesper Böjeryd. Should I stay or should I go? The role of housing in understanding limited inter-regional worker mobility. Working Paper 13/2025, Norges Bank, 2025.

A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427, 2008. doi: 10.1162/rest.90.3.414.

Damian Clarke, Joseph P. Romano, and Michael Wolf. The Romano–Wolf multiple-hypothesis correction in Stata. *The Stata Journal*, 20(4):812–843, 2020. doi: 10.1177/1536867X20976314.

James Feyrer, Erin T. Mansur, and Bruce Sacerdote. Geographic dispersion of economic shocks: Evidence from the fracking revolution. *American Economic Review*, 107(4):1313–1334, April 2017. doi: 10.1257/aer.20151326. URL `https://doi.org/10.1257/aer.20151326`.

Paul Goldsmith-Pinkham, Isaac Sorkin, and Henry Swift. Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8):2586–2624, August 2020. doi: 10.1257/aer.20181047. URL `https://doi.org/10.1257/aer.20181047`.

International Monetary Fund. Norway: 2016 Article IV consultation. IMF Country Report 16/214, International Monetary Fund, 2016.

Grant D. Jacobsen and Dominic P. Parker. The economic aftermath of resource booms: Evidence from boomtowns in the American West. *Economic Journal*, 126(593):1092–1128, 2016.

Ragnar E. Juelsrud and Ella Getz Wold. The importance of unemployment risk for individual savings. *Management Science*, 71(8):6746–6769, 2025. doi: 10.1287/mnsc.2023.00987.

Hege Merete Knutsen, David Jordhus-Lier, Marcin Wojciech Slíwa, and Peder Ressem Østring. The role of unions in shaping work-based precarity on the Norwegian continental shelf. *Economic and Industrial Democracy*, 2025. doi: 10.1177/0143831X251383669. URL `https://doi.org/10.1177/0143831X251383669`. Online first.

Trude Meland. More time off. Industrial Heritage Ekofisk, Norwegian Petroleum Museum, October 2019a. URL `https://ekofisk.industriminne.no/en/more-time-off/`. Accessed 2 October 2026.

Trude Meland. Getting the timing right. Industrial Heritage Ekofisk, Norwegian Petroleum Museum, October 2019b. URL `https://ekofisk.industriminne.no/en/getting-the-timing-right/`. Accessed 2 October 2026.

Atif Mian and Amir Sufi. What explains the 2007–2009 drop in employment? *Econometrica*, 82(6):2197–2223, 2014.

Enrico Moretti. Local multipliers. *American Economic Review: Papers and Proceedings*, 100(2):373–377, 2010.

Matthew J. Notowidigdo. The incidence of local labor demand shocks. *Journal of Labor Economics*, 38(3), 2020.

Offshore Norge. Agreement 129 of 2024: Operator companies (Forbundet Styrke). Collective agreement between the Confederation of Norwegian Enterprise (NHO)/Offshore Norge and the Norwegian Confederation of Trade Unions (LO)/Forbundet Styrke, 2024. URL `https://www.offshorenorge.no/contentassets/2d9953b775474e53a4bd81ce8d4d68a4/avtale-129-2024-eng.pdf`. Accessed 2 October 2026.

Ashesh Rambachan and Jonathan Roth. A more credible approach to parallel trends. *Review of Economic Studies*, 90(5):2555–2591, 2023. doi: 10.1093/restud/rdad018.

Joseph P. Romano and Michael Wolf. Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4):1237–1282, 2005. doi: 10.1111/j.1468-0262.2005.00615.x.

Joseph P. Romano and Michael Wolf. Efficient computation of adjusted p-values for resampling-based stepdown multiple testing. *Statistics & Probability Letters*, 113:38–40, 2016. doi: 10.1016/j.spl.2016.02.012.

David Roodman, Morten Ørregaard Nielsen, James G. MacKinnon, and Matthew D. Webb. Fast and

wild: Bootstrap inference in Stata using boottest. *The Stata Journal*, 19(1):4–60, March 2019. doi: 10.1177/1536867X19830877. URL https://doi.org/10.1177/1536867X19830877.

Pål Schøne and Marte Strøm. Regional oil-shock and household adaption. Working Paper 3765446, SSRN, 2021.

Matthew D. Webb. Reworking wild bootstrap-based inference for clustered errors. *Canadian Journal of Economics*, 56(3):839–858, 2023. doi: 10.1111/caje.12661.

A Offshore work by municipality

Table 10: The municipalities with the most and the least offshore work, 2003

Municipality	Economic region	Employed residents	Working offshore	Share (%)
Panel A. The ten with the most				
Tysnes	Sunnhordland	1,271	54	4.25
Fedje	Bergen	282	11	3.90
Kvinesdal	Flekkefjord	2,617	86	3.29
Bamble	Skien/Porsgrunn	6,638	201	3.03
Farsund	Lyngdal/Farsund	4,391	123	2.80
Bømlo	Sunnhordland	5,051	137	2.71
Lyngdal	Lyngdal/Farsund	3,372	83	2.46
Vega	Brønnøysund	629	15	2.38
Tvedestrand	Arendal	2,614	62	2.37
Sauda	Haugesund	2,290	54	2.36
All 375: p10 0.00, p25 0.10, median 0.28, p75 0.75, p90 1.33 40 at zero; population-weighted mean 0.41				
Panel B. The ten with the least, among municipalities with any				
Gran	Hadeland	6,539	1	0.02
Rakkestad	Fredrikstad/Sarpsborg	3,626	1	0.03
Sørum	Lillestrøm	6,659	2	0.03
Nordre Land	Gjøvik	3,251	1	0.03
Vadsø	Vadsø	3,083	1	0.03
Sel	Nord-Gudbrandsdalen	2,965	1	0.03
Lier	Drammen	11,156	4	0.04
Gjerdrum	Lillestrøm	2,629	1	0.04
Fet	Lillestrøm	4,998	2	0.04
Øyer	Lillehammer	2,459	1	0.04

Note: Residents whose workplace was the continental shelf in the fourth quarter of 2003 as a percentage of employed residents. The sample is the 375 municipalities where at most 0.5 percent of employed residents held onshore oil jobs in 2003, in the municipality or by commuting (Section 3.2). Panel A ends with the distribution of the share across the 375. Forty municipalities had no offshore workers. Nine of them are in Finnmark and six each in Hedmark, Nordland and Trøndelag. The largest, Søndre Land, had 2,692 employed residents. Panel B lists the ten lowest shares among the rest.

B Employment by industry

Table 11: Employed residents by industry of workplace, change per 100 employed residents

	2013–17 (1)	2013–19 (2)	2015–19 (3)
All industries	−1.44***	−1.49**	−0.60*
	(0.38)	(0.52)	(0.33)
Oil and gas, mining and quarrying (05–09)	−0.32***	−0.20***	−0.09
	(0.06)	(0.07)	(0.06)
All other industries	−1.11***	−1.29**	−0.51
	(0.36)	(0.51)	(0.33)
Agriculture, forestry and fishing (01–03)	0.21***	0.23**	0.11
	(0.07)	(0.09)	(0.07)
Manufacturing and mining outside oil (10–33)	−0.53**	−0.59**	−0.40**
	(0.22)	(0.23)	(0.14)
Electricity, water and waste (35–39)	0.00	−0.02	−0.03
	(0.02)	(0.03)	(0.02)
Construction (41–43)	−0.08	−0.11	−0.08
	(0.09)	(0.12)	(0.09)
Trade (45–47)	−0.13	−0.19*	−0.03
	(0.09)	(0.10)	(0.09)
Transport and storage (49–53)	−0.13***	−0.17***	−0.01
	(0.04)	(0.05)	(0.04)
Hotels and restaurants (55–56)	−0.25***	−0.24***	−0.05
	(0.05)	(0.08)	(0.06)
Information and communication (58–63)	−0.02	−0.06	−0.07
	(0.04)	(0.06)	(0.05)
Finance and insurance (64–66)	−0.04	−0.05*	−0.02
	(0.02)	(0.03)	(0.02)
Real estate, professional and technical services (68–75)	−0.15**	−0.17*	−0.11
	(0.05)	(0.08)	(0.07)
Business services (77–82)	−0.05	−0.03	0.08*
	(0.06)	(0.08)	(0.05)
Public administration (84)	0.08	0.05	−0.03
	(0.06)	(0.06)	(0.06)
Education (85)	0.06	0.09	0.04
	(0.06)	(0.10)	(0.08)
Health and social work (86–88)	−0.06	0.05	0.08
	(0.13)	(0.17)	(0.13)
Other services (90–99)	0.00	−0.05	−0.01
	(0.04)	(0.05)	(0.04)
Unspecified	−0.01	−0.02	0.01
	(0.02)	(0.02)	(0.03)

Note: Coefficients per percentage point of offshore work in 2003 in the 375 municipalities of Table 4. The outcome is the change in the number of employed residents whose workplace is in the industry from the fourth quarter of the first year to the fourth quarter of the last, per 100 employed residents in 2013. The data are Statistics Norway's table 13470 by place of residence, with the SIC 2007 groups in parentheses. The register changed source in 2015, so columns (1) and (2) contain the break and column (3) lies inside the new source. Controls, weights, clustering and stars as in column (3) of Table 4.

Table 12: Employed residents in the petroleum-related industries and in manufacturing, change per 100 employed residents

	2013–17 (1)	2013–19 (2)	2015–19 (3)
Petroleum-related industries, Statistics Norway's list	-0.40^{***}	-0.33^{**}	-0.12
	(0.10)	(0.11)	(0.06)
Extraction and support services (06, 09.1)	-0.35^{***}	-0.22^{***}	-0.11^{*}
	(0.05)	(0.07)	(0.06)
Platforms, ships and ship repair (30.11, 33.15)	0.00	-0.04	0.01
	(0.09)	(0.09)	(0.06)
Pipelines, supply vessels and terminals	-0.05^{***}	-0.06^{**}	-0.02
	(0.02)	(0.03)	(0.02)
Manufacturing outside the list	-0.53^{**}	-0.55^{**}	-0.41^{**}
	(0.20)	(0.22)	(0.14)
Machinery and equipment (28)	-0.28^{***}	-0.28^{**}	-0.25^{**}
	(0.11)	(0.11)	(0.11)
Fabricated metal products (25)	-0.12^{**}	-0.15^{***}	-0.08^{*}
	(0.04)	(0.05)	(0.04)
Computer, electronic and optical products (26)	-0.10	-0.09	-0.05
	(0.08)	(0.07)	(0.04)
Chemicals (20)	-0.02	-0.07^{*}	-0.05^{***}
	(0.04)	(0.04)	(0.02)
Ships, boats and other transport equipment (30)	-0.07	-0.10	-0.02
	(0.08)	(0.09)	(0.06)

Note: As Table 11. The petroleum-related industries are Statistics Norway's list (five-digit codes 06, 09.1, 30.111, 30.113, 30.115, 30.116, 33.15, 43.13, 49.5, 50.204, 52.215 and 52.223); manufacturing outside the list is divisions 10–33 less the list's manufacturing codes.

Table 13: Jobs lost for each offshore job lost, with the suppliers purged

	2013–17 (1)	2013–19 (2)	2015–19 (3)
Panel A. Change in employed residents, per 100 employed residents in 2013			
Offshore jobs: extraction and support services (06, 09.1)	-0.35^{***}	-0.22^{***}	-0.11^{*}
	(0.05)	(0.07)	(0.06)
Suppliers: the list's other codes, divisions 25 and 28	-0.45^{***}	-0.53^{***}	-0.34^{***}
	(0.14)	(0.13)	(0.12)
Both, the direct hit	-0.79^{***}	-0.76^{***}	-0.45^{***}
	(0.16)	(0.16)	(0.12)
The rest of the local economy	-0.64^{*}	-0.74	-0.16
	(0.33)	(0.48)	(0.34)
also without the other manufacturing divisions	-0.51	-0.62	-0.08
	(0.29)	(0.46)	(0.34)
All but mining and quarrying (05–09)	-1.11^{***}	-1.29^{**}	-0.51
	(0.36)	(0.51)	(0.33)
All industries	-1.44^{***}	-1.49^{**}	-0.60^{*}
	(0.38)	(0.52)	(0.33)
Panel B. Jobs lost for each offshore job lost			
The rest of the local economy	1.85	3.29	1.50
	(0.92)	(2.30)	(3.39)
	[−0.1, 3.8]	[−1.3, 9.7]	[−14.5, 41.8]
also without the other manufacturing divisions	1.46	2.74	0.75
	(0.87)	(2.23)	(3.32)
	[−0.2, 3.6]	[−1.5, 9.4]	[−20.7, 31.3]
All but mining and quarrying (05–09)	3.21	5.77	4.89
	(0.99)	(2.63)	(4.50)
	[1.3, 5.5]	[1.5, 14.6]	[−17.8, 75.5]

Note: Panel A as Table 11. The suppliers are the petroleum list's codes other than extraction and support services plus fabricated metal products (25) and machinery and equipment (28), the firms described in Section 5.3, which the sample rule does not remove. The rest of the local economy is all employed residents less the direct hit, and the next row also removes the other manufacturing divisions. In 2013 the offshore jobs were 0.7 and the supplier jobs 1.1 per 100 employed residents of the sample. Panel B divides each remainder's coefficient by the offshore coefficient of the same column, with delta-method standard errors from a stacked regression with region clusters in parentheses and 95 percent percentile intervals from a pairs cluster bootstrap over regions (999 draws) in brackets. Over 2015–19 the offshore loss itself is small and imprecise, so the ratios of column (3) are not informative.

Table 14: Jobs lost for each offshore job lost on the supply-chain samples, 2013–17

	Headline (1)	Supply chain ≤ 2% (2)	Supply chain ≤ 1% (3)
Panel A. Change in employed residents 2013–17, per 100 employed residents in 2013			
Offshore jobs: extraction and support services (06, 09.1)	−0.35***	−0.28***	−0.34***
	(0.05)	(0.05)	(0.03)
Suppliers: the list's other codes, divisions 25 and 28	−0.45***	−0.50***	−0.64***
	(0.14)	(0.21)	(0.31)
Both, the direct hit	−0.79***	−0.78***	−0.98***
	(0.16)	(0.21)	(0.30)
The rest of the local economy	−0.64*	−0.91	−0.52
	(0.33)	(0.47)	(0.54)
also without the other manufacturing divisions	−0.51	−0.97*	−0.56
	(0.29)	(0.40)	(0.40)
All but mining and quarrying (05–09)	−1.11***	−1.42**	−1.18*
	(0.36)	(0.48)	(0.59)
All industries	−1.44***	−1.69***	−1.50**
	(0.38)	(0.47)	(0.58)
Panel B. Jobs lost for each offshore job lost			
The rest of the local economy	1.85	3.26	1.53
	(0.92)	(1.96)	(1.60)
	[−0.1, 3.8]	[−0.7, 7.5]	[−1.4, 6.0]
also without the other manufacturing divisions	1.46	3.47	1.65
	(0.87)	(1.78)	(1.19)
	[−0.2, 3.6]	[0.1, 7.7]	[−0.6, 5.6]
All but mining and quarrying (05–09)	3.21	5.08	3.45
	(0.99)	(2.21)	(1.84)
	[1.3, 5.5]	[0.7, 9.9]	[−0.2, 8.5]
Observations	375	290	250

Note: As Table 13, window 2013–17. Column (1) is the headline sample. Columns (2) and (3) keep the headline municipalities whose economic region had at most 2 and at most 1 percent of its 2003 jobs in SN2002 divisions 11 and 35, extraction and the building of ships and platforms, the supply-chain rule of Section 3.2 and Table 2. Table 2 applies the rule to all municipalities; here it is applied inside the headline sample, which is why the counts are 290 and 250 rather than 296 and 256.

C The 2015 change of source in the employment register

Table 15: The 2015 register break by municipality and offshore work

	Mean	Offshore work, 2003	(SE)	Municipalities
Break, all industries	−3.57	−0.23	(0.33)	375
Break, outside mining and manufacturing	−3.74	−0.24	(0.31)	375
Break, public administration, education and health	−1.52	−0.10	(0.38)	375
Change 2014–15, adjusted	0.75	−0.80***	(0.28)	375
Change 2014–15, unadjusted	−2.83	−1.03***	(0.29)	375
Change 2013–14	1.32	0.06	(0.25)	375
Change 2013–17, bridged across the break	5.00	−1.17	(0.67)	375
Change 2013–17, unadjusted	1.42	−1.40**	(0.59)	375
Change 2015–17	2.93	−0.43	(0.38)	375

Note: Employees aged 15–74 by municipality of workplace in the fourth quarter, mapped to the 2019 municipalities, from the two 2015 counts described in Section 6.3: the unadjusted count on the new source and the count adjusted to be comparable with 2014. The break is 100 times the log difference between them. The changes are 100 times log differences, and the bridged 2013–17 change joins 2013 to the adjusted 2015 and the unadjusted 2015 to 2017. Mean is the population-weighted mean in the 375 municipalities of Table 4. The coefficient is on offshore work in 2003 with the controls, weights, inference and stars of column (3) of Table 4.

D Pre-boom placebos

Table 16: Pre-boom outcomes on the 2003 offshore share, by sample

	Population growth		Employment	Registered	
Sample	1980–90	1990–2003	growth 2000–03	unemployment 2000–03	Municipalities
	(1)	(2)	(3)	(4)	
Headline sample	0.93	0.38	0.42	0.02	375
	(0.79)	(0.75)	(0.38)	(0.08)	
Balanced price panel	0.75	0.27	0.31	0.06	271
	(0.82)	(0.79)	(0.38)	(0.08)	
Supply chain at most 2 percent	1.23	1.08	0.63	0.09	290
	(1.11)	(1.21)	(0.61)	(0.13)	
Supply chain at most 1 percent	1.23	2.27	1.28**	−0.02	250
	(1.36)	(1.34)	(0.59)	(0.14)	
Without hub regions, 5 percent	1.24	1.01	0.81	−0.05	272
	(1.18)	(1.06)	(0.52)	(0.12)	
Without hub regions, 3 percent	0.18	1.40	1.09	−0.02	240
	(1.39)	(1.48)	(0.59)	(0.16)	

Note: Coefficients per percentage point of offshore work in 2003. The controls are the regional onshore oil share, log population in 2004 and the 2003 industry shares. The weights are population in 2004. Standard errors clustered by economic region are in parentheses. Stars come from the region wild bootstrap with * $p < 0.10$, ** $p < 0.05$ and *** $p < 0.01$. Population growth is 100 times the log change, employment growth 100 times the log change in employed residents from 2000 to 2003, and registered unemployment the change in percentage points. The samples are the headline sample and the balanced price panel of Table 4, the two supply-chain samples of Table 14 and the headline sample without the hub regions of columns (4) and (5) of Table 6. Municipalities are those of column (3); columns (1), (2) and (4) have one fewer.

E Further checks

Table 17: The bust with and without population weights, 2013–2017

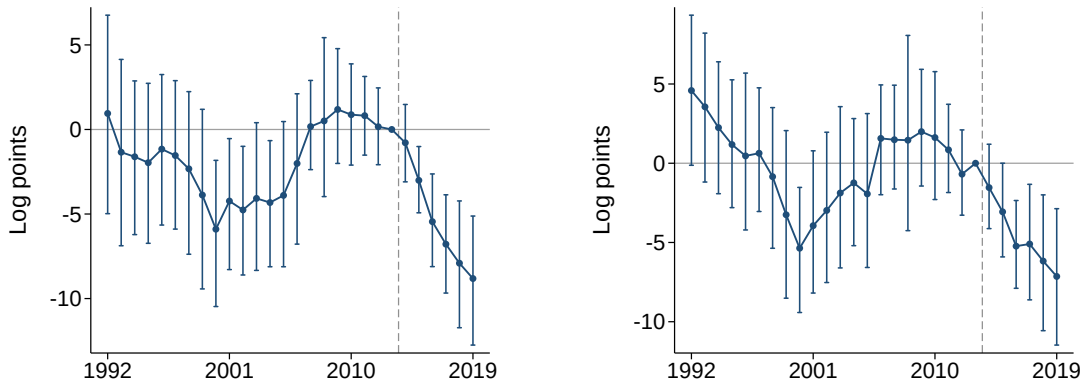
Dependent variable	Income (1)	Employment (2)	Price (3)	Dwelling stock growth (4)	Population (5)
Weighted by population	-1.932^{***}	-1.378^{***}	-7.142^{***}	0.226	−0.284
	(0.385)	(0.386)	(1.444)	(0.302)	(0.272)
Unweighted	-1.305^{***}	-0.688^{**}	-5.643^{***}	0.186	0.180
	(0.338)	(0.279)	(1.649)	(0.277)	(0.210)
Adjusted R^2, weighted	0.35	0.55	0.13	0.12	0.47
Adjusted R^2, unweighted	0.21	0.32	0.06	0.09	0.34
Observations	375	375	271	375	375

Note: Coefficients on offshore work in 2003 with the specification and inference of column (3) of Table 4. The first row is column (3) of Table 4, weighted by population on 1 January 2014, so that the coefficient is the effect for the average resident. The second gives every municipality the same weight, so that it is the effect for the average municipality. Standard errors clustered by economic region in parentheses, stars from the region wild bootstrap.

Table 18: The crisis cushion of 2008 and 2009

	Baseline (1)	Coastal controls (2)
Employment income per adult man	0.63^{***}	0.38^{***}
	(0.14)	(0.12)
Employment income per adult woman	0.26^{*}	0.21
	(0.13)	(0.13)
Median household income after tax	0.24	0.16
	(0.13)	(0.11)
Registered unemployment rate	−0.01	0.02
	(0.04)	(0.04)
Average sale price	0.88	0.34
	(1.53)	(1.51)
Detached-house price per m^2	1.87^{*}	1.16
	(0.82)	(1.02)
Apartment price per m^2	4.47^{***}	5.22^{***}
	(1.02)	(1.45)
Number of free-market sales	0.71	2.02
	(1.33)	(1.80)

Note: Each entry is $(\beta_{2008} + \beta_{2009})/2 - (\beta_{2006} + \beta_{2007} + \beta_{2010} + \beta_{2011})/4$, the average event-study coefficient of 2008 and 2009 relative to the two years on either side. It is per percentage point of offshore work in 2003 and comes from event studies with base year 2013 on the headline sample from 2005, as in Figure 3. Incomes and prices are in log points, registered unemployment in percentage points. Column (2) adds the coastal controls of Section 3.4. Standard errors clustered by economic region in parentheses and stars from the region wild bootstrap as in Table 4. The detached and apartment prices per square metre are observed where Statistics Norway publishes them.



(a) Baseline controls **(b)** Coastal controls

Figure 5: The average sale price from 1992, relative to 2013

Note: Coefficients on offshore work in 2003 by year for 100 times the log difference of the average sale price from 2013, with 95 percent region wild bootstrap intervals. The sample is the headline municipalities with a price in every year from 1992 to 2019 (Statistics Norway table 06726). Panel (b) adds the coastal controls of Section 3.4. A joint test that the coefficients of 1992 to 2002, with base year 2003, are zero has a p-value of 0.30 in panel (a) and 0.05 in panel (b).

Table 19: The 2003 share, the 2013 share and the instrumented 2013 share, bust 2013–2017

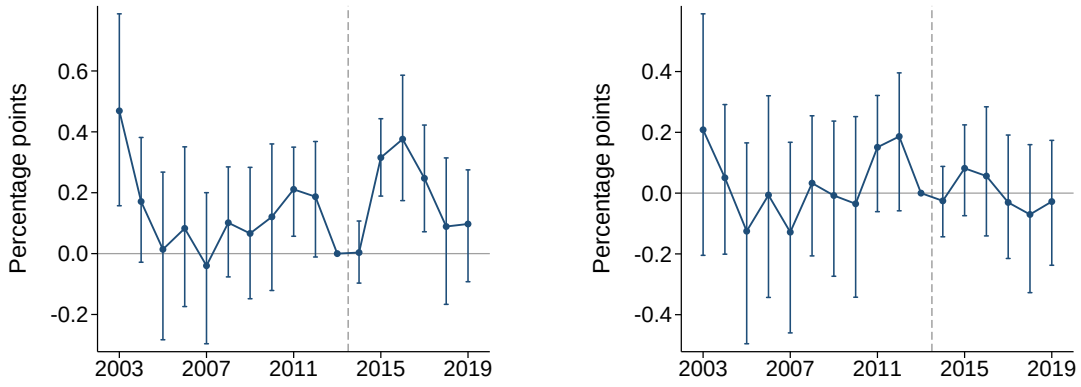
	Income (1)	Employment (2)	Price (3)
Offshore work, 2003	-1.932^{***}	-1.378^{***}	-7.142^{***}
	(0.385)	(0.386)	(1.444)
Offshore work, 2013, least squares	-1.385^{***}	-0.810^{***}	-5.174^{***}
	(0.259)	(0.290)	(1.244)
Offshore work, 2013, instrumented	-1.712^{***}	-1.221^{***}	-6.400^{***}
	(0.374)	(0.388)	(1.435)
Observations	375	375	271

Note: Coefficients per percentage point of offshore work, with the controls, weights and inference of column (3) of Table 4. The first row is column (3) of Table 4 and the second regresses the same outcomes on the 2013 share by least squares. The third instruments the 2013 share by the 2003 share, with stars from the wild restricted efficient bootstrap over regions. The first stage is 1.13 with a Kleibergen–Paap F of 342 (337 on the price sample), which says only that the share is persistent.

Table 20: The bust with the centrality classifications of Statistics Norway, 2013–2017

	Income (1)	Employment (2)	Price (3)
Baseline	-1.932^{***}	-1.378^{***}	-7.142^{***}
	(0.385)	(0.386)	(1.444)
Distance to the nearest large city	-1.824^{***}	-1.106^{***}	-6.219^{***}
	(0.328)	(0.345)	(1.450)
Centrality classes, 2008	-2.008^{***}	-1.213^{**}	-7.099^{***}
	(0.383)	(0.454)	(1.532)
Centrality groups, 2018	-1.962^{***}	-1.062^{**}	-6.340^{***}
	(0.327)	(0.394)	(1.584)
Observations	375	375	271

Note: Coefficients on offshore work in 2003 with the specification, weights and inference of column (3) of Table 4. The second row adds the distance to the nearest of Oslo, Bergen, Trondheim and Stavanger as in Table 7. The third adds the 2008 centrality standard of Statistics Norway as three class dummies and the fourth its 2018 centrality index as five group dummies.

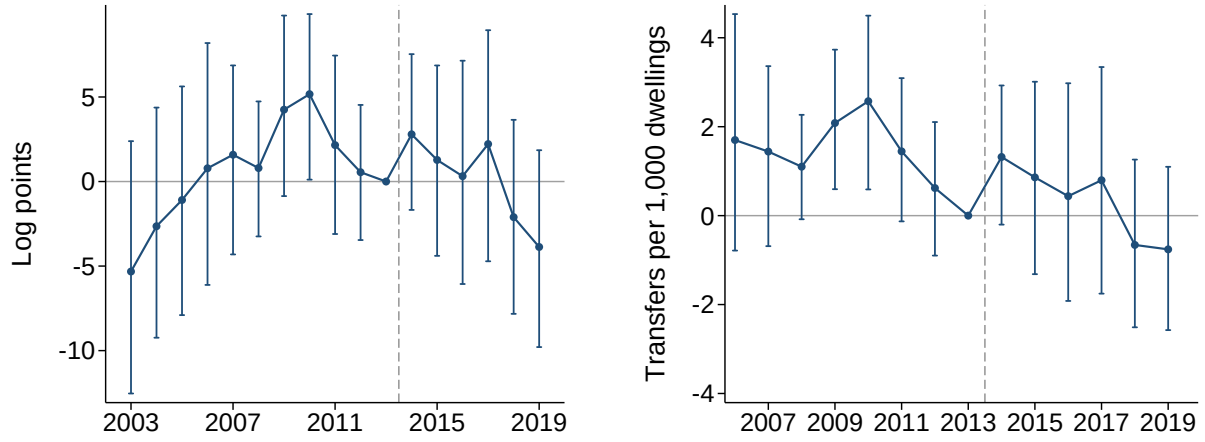


(a) Baseline controls **(b)** Coastal controls

Figure 6: Registered unemployment, 2003–2019

Note: Coefficients on offshore work in 2003 by year for the change in the registered unemployment rate from November 2013 in percentage points, with 95 percent region wild bootstrap intervals, on the headline sample. Panel (b) adds the coastal controls of Section 3.4.

F Sales volumes



(a) Number of free-market sales, log points

(b) Sales per 1,000 dwellings

Figure 7: Offshore work and the number of dwelling sales, 2003–2019

Note: As Figure 3 in calendar years, for the number of free-market sales and for sales per 1,000 dwellings. Sales are 100 times the log difference from 2013. Sales per 1,000 dwellings are the difference from their 2013 value. The second series starts in 2006 with the dwelling stock series. The sample is the 269 municipalities with at least 20 free-market sales in 2013.

G Data sources

The supply-chain rule of 2003. Statistics Norway's list of petroleum-related industries (5-digit SN2007, table 13470) starts in 2008. Its correspondence table maps the list into SN2002 as follows: extraction and its support services (06, 09.1) into division 11; the building of ships, platforms and modules, installation work on them and ship repair (30.111, 30.113, 30.115, 30.116, 33.15) into division 35; test drilling (43.13) into 45.120; pipeline transport (49.5) into 60.300; offshore supply shipping (50.204) into 61.106; and supply bases (52.223) into 63.224. Services for the operation of pipelines (52.215, 136 jobs in 2008) have no entry. Of the list's 79,381 jobs located in municipalities in 2008, 90 percent fall into codes that map to divisions 11 and 35. Their share of a region's jobs in 2003 correlates 0.99 with the same share in 2008. Division 35 also contains small boats, leisure boats, rail and aircraft, so the 2003 rule is somewhat broader than the list.

Construction checks. Every analysis dataset is rebuilt from the raw downloads by the R scripts in the project, and a rebuild reproduces them exactly. The checks in `R/11_sanity_checks.R` confirm the following.

- Every national total reaches the 2019 map exactly, except that 22,000 jobs whose place of work is the shelf or unknown have no municipality in the place-of-work register.
- No municipal code with data is lost in the mapping.

- The event-study panel equals the long differences at every point where they measure the same change.

Table 21: Public data sources

Table	Content	Years used
Statistics Norway (StatBank), municipality level		
03311	Employed persons by broad SN2002 industry group (including division 11), by residence and by place of work	2003
03321	Employed persons by municipality of work and of residence (continental shelf)	2000–2019
08545	Employed persons by SN2002 two-digit industry, by place of work (shelf; supply chain, divisions 11 and 35), by residence (fishing, transport equipment, water transport) and by sex (shelf)	2000, 2003, 2008
13470	Employed persons by 5-digit SN2007 industry, place of work (petroleum-related list) and place of residence (industry of workplace, suppliers)	2008, 2013–2019
11365	Employees by place of work and 17 industry groups, 2015 adjusted for the change of source	2013–2015
11396	Employees by place of work and 17 industry groups, unadjusted	2015–2019
06944	Household income after tax and total, median, all households	2005–2019
11607	Employed persons 20–66 by immigrant background	2001–2019
11616	Employed residents 15–74	2000–2003
10540	Registered unemployed 15–74 (NAV), per cent	2000–2019
13563	Residents 15–74 by labour-market status	2008–2019
06913	Population on 1 January, births, deaths, growth	1970–2019
06265	Dwellings by building type, 1 January	2006–2019
05940	Dwellings started and completed	2000–2019
06035	Price per m^2 and number of sales, by dwelling type	2002–2019
06726	Free-market sales of dwellings, number and value	2000–2019
03068	Average income from employment and gross income per resident aged 17 and over, by sex (tax records)	2005–2019
12314	Employees, jobs and earnings by industry division (88 groups, SIC2007), national	2016
08050	Planned and actual petroleum investment (national)	2002–2019
04170	Accrued petroleum investment, older series (equal to 08050 where both exist)	2000–2019
14701	Consumer price index, annual	2000–2019
Statistics Norway, other		
Census	1980 Population and Housing Census, municipal booklets, table 9 (employed persons by place of work; row "North Sea")	1980
Statistics Norway, classifications (KLASS)		
131	Municipality code changes	1970–2019
108	Economic regions (2018 version, 2019 map)	2019
Corr. 108	Correspondence table SN2007–SN2002 (a different series of numbers from the classifications)	2009
Norwegian Offshore Directorate (FactPages and FactMaps)		
Fields	Outlines of the 139 oil and gas fields	current
Other		
Eurostat GISCO	County boundaries (NUTS 2016 level 3)	2016
FRED (EIA)	Brent spot price, daily (DCOILBRENTEU)	2000–2019
Wikidata	Municipal administrative centres	current
Heliports	Coordinates of the five offshore heliports in use in 2003	2003